MAXIMAL CONVERGENCE ABSCISSAS FOR GENERAL FREQUENCIES

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AI assistance was used in adapting and checking the argument and the exposition.

In arXiv:1902.02073v1, Defant and Schoolmann ask: "Given a frequency λ, give reasonable upper and lower estimates for $\sup_{D\in H_p(\lambda)}\sigma_c(D)$ and $\sup_{D\in H_p(\lambda)}\sigma_c(D)$, for example in terms of $L(\lambda)$." This is Problem 4.17, p. 21; the published Problem 4.21 corrects the second σ_c to σ_a.

For every frequency, with $L = L(\lambda)$,

$$L/2 \le C_p \le S_p \le L/p \quad (1 \le p \le 2), \qquad L/p \le C_p \le S_p = L/2 \quad (2 \le p < \infty).$$

Here $L(\lambda) = \limsup_n (\log n)/\lambda_n$, and $\sigma_c(D)$ and $\sigma_a(D)$ are the abscissas of ordinary and absolute convergence; write $C_p = \sup_{D\in H_p(\lambda)}\sigma_c(D)$ and $S_p = \sup_{D\in H_p(\lambda)}\sigma_a(D)$.

The source is Defant and Schoolmann, *Hardy spaces of general Dirichlet series—a survey*, arXiv:1902.02073v1.

These universal bounds are sharp.

Answered: sharp bounds in terms of $L(\lambda)$ and p. Evidence: Sketch.

Proof. For finite L, choose L' with $L < L' < \tau$. Since $\limsup_{n\to\infty}(\log n)/\lambda_n = L$, we have $\lambda_n > (\log n)/L'$ eventually, and hence

$$\sum_n e^{-\tau\lambda_n} < \infty.$$

If $L = \infty$, the upper bounds in the theorem are automatic. For $D = \sum a_n e^{-\lambda_n s} \in \mathcal{H}_p(\lambda)$, the coefficient map satisfies $\|a\|_{\ell^\infty} \le \|D\|_1$, and orthogonality gives $\|a\|_{\ell^2} \le \|D\|_2$. For $1 \le p \le 2$, interpolation yields $\|a\|_{\ell^{p'}} \le \|D\|_p$; for $p \ge 2$, monotonicity of the L^p norms gives $\|a\|_{\ell^2} \le \|D\|_p$. Hölder's inequality and Cauchy–Schwarz now give $S_p \le L/p$ when $p \le 2$ and $S_p \le L/2$ when $p \ge 2$. Since convergence implies absolute-convergence abscissa is no smaller, $C_p \le S_p$.

If $L = 0$, the series $D_0(s) = \sum_n n^{-2}e^{-\lambda_n s}$ converges in $\mathcal{H}_p(\lambda)$ because each character has norm one. Since $\lambda_n/\log n \to \infty$, for every real $\sigma < 0$ its terms at σ tend to infinity. Thus $C_p \ge 0$, giving the lower bound in this case.

Suppose $L > 0$ and fix $0 < M < L$. Choose N_k recursively so that $N_k \ge e^k$, $N_k \ge 2N_{k-1}$, and $0 < \lambda_{N_k} < (\log N_k)/M$. Such indices exist by the definition of L. The disjoint blocks $B_k = \{n : N_k/2 < n \le N_k\}$ have $|B_k| \ge N_k/2$ and $N_k \le 2|B_k|$. Monotonicity of λ gives, for $\sigma, b \ge 0$,

$$\sum_{n\in B_k} be^{-\sigma\lambda_n} \ge b|B_k|e^{-\sigma\lambda_{N_k}} \ge b|B_k|^{1-\sigma/M}2^{-\sigma/M}.$$

Put $\theta = \max\{1/2, 1-1/p\}$ and $Q_k = \sum_{n\in B_k} h_{\lambda_n}$. Orthogonality and the triangle inequality give $\|Q_k\|_2 = |B_k|^{1/2}$ and $\|Q_k\|_\infty \le |B_k|$, so $\|Q_k\|_p \le |B_k|^\theta$: use $\|Q_k\|_p \le \|Q_k\|_2$ for $p \le 2$, and $\|Q_k\|_p^p \le \|Q_k\|_\infty^{p-2}\|Q_k\|_2^2$ for $p \ge 2$. Therefore

$$D_1(s) = \sum_k k^{-2}|B_k|^{-\theta} \sum_{n\in B_k} e^{-\lambda_n s}$$

converges in $\mathcal{H}_p(\lambda)$. If $\sigma < (1-\theta)M$, put $\varepsilon = 1-\theta-\sigma/M > 0$. The sum of the terms of $D_1(\sigma)$ over B_k is at least $c_\sigma k^{-2}|B_k|^\varepsilon$, which tends to infinity because $|B_k| \ge e^k/2$. These

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are differences of partial sums, so D_1 diverges at σ. Letting $M \uparrow L$ proves $C_p \geq (1-\theta)L = \min\{L/2, L/p\}$.

For the absolute-convergence lower bound, Khintchine's inequality and Fubini's theorem give signs ε_n, $n \in B_k$, such that $\|\sum_{n\in B_k} \varepsilon_n h_{\lambda_n}\|_p \leq K_p |B_k|^{1/2}$. Hence

$$D_2(s) = \sum_k k^{-2}|B_k|^{-1/2} \sum_{n\in B_k} \varepsilon_n e^{-\lambda_n s}$$

converges in $\mathcal{H}_p(\lambda)$. If $\sigma < M/2$, the absolute sum over B_k is at least $c_\sigma k^{-2}|B_k|^{1/2-\sigma/M}$, which diverges as $k \to \infty$. Thus $S_p \geq M/2$, and then $S_p \geq L/2$ by letting $M \uparrow L$.

For sharpness, fix $A, B > 0$. Let $M_j = \lceil e^{Aj} \rceil$. Choose $\delta_j > 0$ so that $M_j\delta_j < 1/4$ and $\{1, \delta_1, \delta_2, \ldots\}$ is linearly independent over $\mathbb{Q}$. The frequency

$$\mu_{j,k} = j + k\delta_j, \qquad j \geq 1, \quad 0 \leq k < M_j,$$

is increasing when ordered by blocks; the blocks are disjoint and contained in $[j, j+1/4)$. Their cumulative counting function gives $L(\mu) = A$.

The corresponding characters can be written $z_0^j z_j^k$ on a product torus, so a polynomial in $\mathcal{H}_p(\mu)$ has the form $F = \sum_j z_0^j P_j(z_j)$, with $\deg P_j < M_j$. Conditional expectation onto the coordinates z_0, z_j, followed by extraction of the jth Fourier coefficient in z_0, is contractive on L^p and gives

$$\|P_j\|_{L^p(\mathbb{T})} \leq \|F\|_{\mathcal{H}_p(\mu)}.$$

This remains true by density for every $F \in \mathcal{H}_p(\mu)$.

Let $D_m(z) = 1 + \cdots + z^{m-1}$. The geometric-sum bound $|D_m(e^{it})| \leq \min(m, C/|t|)$ gives $\|D_m\|_1 \leq C\log(m+1)$ and $\|D_m\|_p \leq C_p m^{1-1/p}$ for $p > 1$. If Q is any initial segment of the coefficients of a polynomial P of degree less than m, convolution with a Dirichlet kernel and Young's inequality give $\|Q\|_p \leq C\log(m+1)\|P\|_p$. Nikolskii's inequality then yields $\|Q\|_\infty \leq C_p m^{1/p}\log(m+1)\|P\|_p$ for $p > 1$. At $p = 1$, Hausdorff–Young bounds each coefficient of P by $\|P\|_1$, so $\|Q\|_\infty \leq m\|P\|_1$. Consequently every partial sum in block j is bounded on $\operatorname{Re} s \geq \sigma$ by

$$C_p e^{-j\sigma} M_j^{1/p} \log(M_j+1)\|F\|_{\mathcal{H}_p(\mu)}$$

for $p > 1$, and by $e^{-j\sigma}M_j\|F\|_{\mathcal{H}_1(\mu)}$ for $p = 1$. Summing over j proves uniform convergence on every such half-plane when $\sigma > A/p$, hence $C_p(\mu) \leq A/p$.

For the reverse inequality set $K_{p,j} = \|D_{M_j}\|_{L^p(\mathbb{T})}$ and take

$$F_p(s) = \sum_{j\geq 1} j^{-2} K_{p,j}^{-1} \sum_{k=0}^{M_j-1} e^{-(j+k\delta_j)s}.$$

The jth block has $\mathcal{H}_p(\mu)$ norm j^{-2}, so the series converges in $\mathcal{H}_p(\mu)$. Its coefficients are positive. At real $0 \leq \sigma < A/p$, its jth block value is at least

$$j^{-2}K_{p,j}^{-1} M_j e^{-(j+M_j\delta_j)\sigma} \geq c_\sigma j^{-2} K_{p,j}^{-1} M_j e^{-j\sigma}.$$

For $p > 1$, $\|D_{M_j}\|_p \leq C_p M_j^{1-1/p}$, so this lower bound grows exponentially like $e^{j(A/p-\sigma)}$ up to a polynomial factor. For $p = 1$, $\|D_{M_j}\|_1 \leq C\log(M_j+1)$, and it grows like $e^{j(A-\sigma)}$ up to a polynomial factor. Thus the partial sums fail to converge for every $\sigma < A/p$, proving $C_p(\mu) = A/p$.

Next choose a positive increasing sequence ν_n in small disjoint intervals about $\log(n+1)/B$, avoiding the rational span of $\{1, \delta_1, \delta_2, \ldots\}$ and all previously chosen terms. Then ν_n are rationally independent, $\operatorname{span}_{\mathbb{Q}}\{\nu_n\} \cap \operatorname{span}_{\mathbb{Q}}\{1, \delta_j : j \geq 1\} = \{0\}$, and $L(\nu) = B$. Its characters are independent uniform circle variables. For a finite first-chaos sum $X = \sum_n a_n z_n$,

$$\|X\|_2^2 = \sum_n |a_n|^2, \qquad \|X\|_4^4 = 2\Big(\sum_n |a_n|^2\Big)^2 - \sum_n |a_n|^4 \leq 2\Big(\sum_n |a_n|^2\Big)^2.$$

Interpolation gives $\|X\|_1 \geq \|X\|_2^3/\|X\|_4^2 \geq 2^{-1/2}\|a\|_2$. For each integer $r \geq 1$, expansion of the $2r$th moment also gives $\|X\|_{2r}^{2r} \leq r!\|a\|_2^{2r}$. Thus the $\mathcal{H}_p(\nu)$ norm is equivalent to the ℓ^2 norm of the coefficients for every finite p. Cauchy–Schwarz gives $C_p(\nu) \leq B/2$, and the coefficient sequence $a_n = ((n+1)^{1/2}\log(n+1))^{-1}$ belongs to ℓ^2 and has convergence abscissa $B/2$, so $C_p(\nu) = B/2$.

Let λ be the sorted union of μ and ν. Its counting function is the sum of the two component counting functions, so $L(\lambda) = \max\{A, B\}$. The direct-sum condition realizes $\mathcal{H}_p(\lambda)$ on independent product factors. Conditional expectation onto either factor extracts the corresponding component contractively. If $\sigma > \max\{A/p, B/2\}$, the μ component converges uniformly by the block estimate above and the ν component converges absolutely by Cauchy–Schwarz. A partial sum in the sorted union consists of prefixes of these two components, so $C_p(\lambda) \leq \max\{A/p, B/2\}$. The positive witnesses in the two factors give the reverse inequality.

For $1 \leq p \leq 2$, the general upper estimate for $S_p(\mu)$ gives $\sigma_a(F_\mu) \leq A/p$ for every μ component, while the ν component has absolute convergence abscissa at most $B/2$. Hence $S_p(\lambda) \leq \max\{A/p, B/2\} = C_p(\lambda)$, and equality follows from $C_p \leq S_p$. For $1 \leq p \leq 2$, choosing $B = L$ and $A = pc$ realizes $C_p = S_p = c$ for every $L/2 \leq c \leq L/p$. For $2 \leq p < \infty$, choosing $A = L$ and $B = 2c$ realizes $C_p = c$ for every $L/p \leq c \leq L/2$; the general theorem gives $S_p = L/2$. Thus all the bounds are attained. □